Ranjani Narayanan

Email : rnarayanan35@gatech.edu
https://www.linkedin.com/in/ranjanin2 | OrciD: 0000-0003-4351-6501 | Mobile : +1 (908)-340-7845
Portfolio: ranjanin.com

Education

- **Doctor of Philosophy (Ph.D.) in Electrical and Computer Engineering** — Aug 2021 – 2026 (Expected)
  Georgia Institute of Technology, Atlanta
  Dissertation: *Investigating a human-centered approach towards supporting Shared Mental Models in hierarchical human-agent teams for decision making.*
  Advisor: Prof. Karen Feigh
- **Master of Science (MS) in Electrical Engineering** — Aug 2019 – May 2021
  University of Pennsylvania, Philadelphia — ***GPA:*** *3.8/4.0*
- **Bachelor of Technology (B.Tech) in Electrical Engineering** — Aug 2015 – May 2019
  Sardar Patel College of Engineering, Mumbai — ***GPA:*** *9.3/10.0*

Research Experience

- **Graduate Research Assistant- Georgia Institute of Technology** — Atlanta, GA
  Advised by Prof. Karen M. Feigh, Cognitive Engineering Center — *Feb 2022 - Present*
  - Designed and executed IRB-compliant, controlled human-in-the-loop studies to **investigate the causal effects of AI-driven interventions on user behavior and performance**. Applied human factors and cognitive engineering methods, including heuristic evaluations and cognitive walkthroughs, to **iteratively develop and refine research hypotheses.**
  - Developed and maintained **multi-modal human behavioral data collection and processing** pipelines.
  - Created and validated **human-centered evaluation metrics** to measure **quantitative human-in-the-loop** performance, decision strategies, decision effectiveness, speed, etc. along with socio-technical factors such as trust calibration, reliance behaviors, workload, and **qualitative** subjective perception of AI-assistance.
  - Utilized **statistical inference, mixed-effects modeling, and regression analyses** to **evaluate behavioral outcomes and team-level performance impacts** of AI system designs.
  - **Translated empirical findings into clear, actionable insights** that informed the design and improvement of AI-assisted workflows and decision-support systems.
  - **Published 8+ peer-reviewed papers** in leading international journals, conferences, and books contributing to interdisciplinary research at the intersection of cognitive science, human factors, human-AI interaction, and AI.
  - **Co-led the development of a successful three-year research grant proposal**; communicated project plans and progress through written reports and presentations to stakeholders at the Office of Naval Research on annual and biannual cycles.

Industry Experience

- **GE Aerospace Research** — Niskayuna, NY
  Research Intern — *June - Aug 2025*
  - Created a **multi-domain evaluation dataset** (Engineering and Scientific Q&A) to **assess text completeness** in **retrieval-augmented generation (RAG) systems**, **benchmarking across** 2 LLMs, 2 embedding models, and multiple temperature settings.
  - Developed a domain-agnostic proof-of-concept pipeline to **improve LLM trustworthiness** by evaluating three dimensions: (1) **response completeness and hallucination detection**, (2) **knowledge sufficiency** for answering queries, and (3) **query completeness** relative to available knowledge.
  - Implemented **user-facing diagnostics** that highlight contradictions, missing facts, partial answers, and hallucinations directly within model responses and retrieved knowledge, **enabling early identification of unreliable outputs.**
  - **Communicated findings** at a GE Research Center–wide poster session (1,000+ attendees comprising **technical and non-technical audience**) and collaborated with GE Research Bangalore and the VP of Engineering at GE Aerospace HQ to **inform downstream model improvement and cross-domain adoption.**

- **GE Aerospace Research** Niskayuna, NY
  Research Intern *June - Aug 2024*
  - Prototyped a **domain-agnostic, automated NER pipeline** that uses an LLM (Instruct model) for **synthetic dataset creation** and BERT-based models for **low-latency training and inference**. Demonstrated recognition of 11+ different entity types using **prompt engineering** for constructing a synthetic dataset and **heuristic filtering** for **robust labeling** of LLM outputs. Reduced dependency on LLM inference at runtime, enabling scalable, low-latency entity extraction across domains through fine-tuned BERT models.
  - Co-authored a conference publication for **post-training evaluation of open-source LLMs** on **2 newly curated datasets** (Arithmetic and Interval Labeling tasks), spanning model scales from 3.8B to 236B parameters. Designed **synthetic benchmarks** with semantic-preserving perturbations to systematically assess model degradation on math problems.

- **Autodesk**
  Machine Learning Intern *May - Aug 2022*
  - Extracted and structured user-stored features to **enhance AutoCAD's recommendation system**, applying NLP methods to **parse metadata from 1M+ active users** and producing a clean unsupervised dataset of 55K high-quality data samples.
  - Performed **feature engineering** and generated **high-dimensional BERT embeddings for clustering**, increasing effective data coverage from 23% to 92% and enabling **robust insights from large-scale user behavior in production settings**.

Journal Publications

- **Narayanan, R.** & Feigh, K. (2026). Designing for Oversight: An Empirical Investigation of the Dual Impact of AI Dependency and Information Abstraction on Human Supervision in Decision-Making Teams. *International Journal on Human Computer Interaction,* 1–30.doi:10.1080/10447318.2026.2618568
- Walsh, S. E., **Narayanan, R.**, & Feigh, K. (under review, 2025). The Role of Shared Mental Models in AI-advised Decision Support. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*

Conference Publications

- **Narayanan, R.**, Cohen, M., Feigh, K., & Cooke, N. Two Sides of the Same Coin? Joint Perspectives from Shared Mental Models and Interactive Team Cognition Theories on Human-AI Team Cognition. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 2025.* 69(1), 412-417. doi:10.1177/10711813251358788
- Peshave, A., Hossain, K., Kubricht, J., **Narayanan, R.**, Burpee, Z., & Agarwal, A. (accepted, 2025). Evaluating Large Language Models on Arithmetic and Interval Labeling Problems with Syntactic Perturbations. *2025 IEEE International Conference on Data Mining Workshops (ICDMW).*
- **Narayanan, R.** & Feigh, K. Human Assessment of AI Errors and its Impact on Hybrid Teaming for Decision Making. *2025 IEEE Conference on Cognitive and Computational Aspects of Situation Management (CogSIMA), Duisburg, Germany, 2025, pp. 103-110. doi:10.1109/CogSIMA64436.2025.11079477* ***(Winner of Best Student Paper Award)***
- **Narayanan, R.** and Feigh, K. M. (2025). Impact of Team Models in Hierarchical Human-Agent Decision-Making Teams. In Proceedings of the 20th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications - Volume 1: GRAPP, HUCAPP and IVAPP, ISBN 978-989-758-728-3, ISSN 2184-4321, pages 452-463. doi:10.5220/0013097400003912 **(Nominated for Best Paper & Best Student Paper Awards)**
- **Narayanan, R.,** Walsh, S. E., & Feigh, K. M. (2023). Development of Mental Models in Decision-Making Tasks. *Proceedings of the 67th Human Factors and Ergonomics Society Annual Meeting.* doi:10.1177/21695067231192195

Book Chapters

- **Narayanan, R.** & Feigh, K. (accepted, 2025). How well do we rely on reliance? On the under-utilization of reliance-based metrics towards studying human response to automation assistance. *Advancements in Human Agent Teaming Research Infrastructure: Testbeds, Metrics, and Concepts. Editors Erin K. Chiou, Douglas S. Lange, Jason H. Wong, Julie Marble. CRC Press, Taylor & Francis.*

Workshop Proceedings

- **Narayanan, R.,** & Feigh, K. M. Influence of Human-AI Team Structuring on Shared Mental Models for Collaborative Decision Making. *In Proceedings of Workshop on Theory of Mind in Human-AI Interaction at* ***ACM CHI 2024***.

Technical Skills

- **Programming:** Python, R, MATLAB
- **Tools:** Pytorch, Tensorflow, Keras, Huggingface, OpenCV, Scikit-learn, Pandas, Spacy, SciPy, OpenAI Gym, Git
- **Big Data and Cloud:** AWS (EC2, S3, Bedrock)
- **Simulation & Other Toolkits:** Simulink (MATLAB), VREP
- **Research Methods in HCI:** Statistical modeling & large-scale data analysis, Causal Inference, Human-subjects Experimental Design, Data Visualization, Survey and Interview design, Mixed-effects models, regression tools, Mixed-methods (quantitative & qualitative analytics)
- **AI/ML:** Machine Learning (supervised and unsupervised learning), Deep Learning, Computer Vision, Generative AI (LLMs, VAEs, GANs), Reinforcement Learning, Natural Language Processing, Transfer Learning, Model fine tuning

Honors & Awards

- **IEEE CogSIMA 2025 - Best Student Paper Award** For the conference paper on "Human Assessment of AI Errors and its Impact on Hybrid Teaming for Decision Making".
- **VISIGRAPP 2025 - Nominated for Best Paper & Best Student Paper Awards** For the conference paper on "Impact of Team Models in Hierarchical Human-Agent Decision Making Teams".
- **Georgia Tech Student Government Association (2025):** Travel Grant ($1000).
- **Career, Research, and Innovation Development Conference (CRIDC) Poster Competition Winner, Georgia Institute of Technology (2024)** For my poster on Shared Mental Models for Human-AI Teaming.
- **INSPIRE Fellowship (2015):** Government of India.

Reviewing

- **Book:** Advancements in Human Agent Teaming Research Infrastructure: Testbeds, Metrics, & Concepts.
- **Journals:** International Journal of Human Computer Interaction (Student co-reviewer with Prof. Karen Feigh)
- **Journals:** International Journal of Industrial Ergonomics (Student co-reviewer with Prof. Karen Feigh)
- **Conferences:** ACM Conference on Human Factors in Computing Systems (ACM CHI), Human Factors and Ergonomics Society (HFES)

Activities and Professional Service

- **AI Safety Fellowship at Georgia Institute of Technology (Fall 2025):** Successfully completed a fellowship program on AI Safety and Governance. Analyzed emerging AI governance frameworks and safety standards to propose mitigation strategies for catastrophic risks associated with frontier models. Synthesized complex AI safety literature into actionable policy recommendations regarding model transparency and auditability.
- **Contributor in Social Action Research at Georgia Institute of Technology (Fall 2025):** Successfully completed a social action research program involving deep inquiry, qualitative observation, and real-world evaluation. Collaborated with a research team to identify opportunities to enhance the graduate student experience at Georgia Tech and developed actionable recommendations aimed at creating positive campus-level social impact.
- **Conference Session Chair** for presentation track "Innovations in Research Methods" at the ASPIRE, 69th International Annual Meeting of the Human Factors and Ergonomics Society (HFES), Chicago, USA 2025.
- **Panel Discussion Moderator** ASPIRE, 69th International Annual Meeting of the Human Factors and Ergonomics Society (HFES), Chicago, USA 2025.
- **Alumni Interviewer, University of Pennsylvania:** Interviewer for incoming candidates of undergraduate students for academic years 2021-22 & 2022-23

References (Thesis Committee)

- **Prof. Karen Feigh, Georgia Tech (PhD Advisor):** karen.feigh@gatech.edu
- **Prof. Nancy J Cooke, Arizona State University:** nancy.cooke@asu.edu
- **Prof. Samuel Coogan, Georgia Tech:** sam.coogan@gatech.edu
- **Dr. Zahra Ashktorab, Microsoft:** zashktorab@gmail.com